

Appendix

A.1 Raw Cross-Sectional Areas (mm²) (Standard Deviation in Parentheses) Measured Directly From MRI Scans

Muscles	VERTEBRAL LEVEL					
	L5	L4	L3	L2	L1	T12
R rectus abdominis	787 (250)	750 (207)	670 (133)	712 (239)	576 (151)	
L rectus abdominis	802 (247)	746 (181)	693 (177)	748 (240)	514 (99)	
R external oblique		915 (199)	1,276 (171)	1,158 (222)		
L external oblique		992 (278)	1,335 (213)	1,351 (282)		
R internal oblique		903 (83)	1,515 (317)	1,055 (173)		
L internal oblique		900 (115)	1,424 (310)	1,027 (342)		
R trans. abdominis	119 (22)	237 (82)	356 (110)	596 (50)		
L trans. abdominis	175 (57)	224 (48)	376 (115)	646 (183)		
R abdominal wall*	1,104 (393)	2,412 (418)	3,269 (422)	3,051 (463)		
L abdominal wall*	1,146 (377)	2,420 (475)	3,329 (468)	3,111 (556)		
R longissimus thor.			747 (162)	1,175 (370)	1,248 (228)	1,095 (222)
L longissimus thor.			782 (129)	1,089 (251)	1,180 (184)	1,258 (347)
R iliocostalis lumb.			1,368 (341)	1,104 (181)	1,181 (316)	921 (339)
L iliocostalis lumb.			1,395 (223)	1,150 (198)	1,158 (247)	835 (400)
R multifidus			447 (271)	343 (178)	290 (96)	289 (66)
L multifidus			472 (269)	366 (157)	324 (95)	312 (76)
R latissimus dorsi			232 (192)	429 (202)	717 (260)	1,014 (264)
L latissimus dorsi			256 (217)	372 (161)	682 (260)	960 (310)
R erector mass**	905 (331)	2,151 (539)	2,831 (458)	2,854 (547)	2,615 (405)	2,614 (584)
L erector mass**	986 (338)	2,234 (476)	2,933 (382)	2,833 (456)	2,723 (428)	2,601 (559)
R psoas	1,606 (198)	1,861 (347)	1,594 (369)	1,177 (285)	513 (329)	330 (210)
L psoas	1,590 (244)	1,820 (272)	1,593 (291)	1,211 (298)	488 (250)	462 (190)
R quadratus lumb.		725 (209)	701 (212)	552 (192)	392 (249)	320 (197)
L quadratus lumb.		625 (249)	746 (167)	614 (189)	404 (220)	326 (5)
Disc area	1,360 (276)	1,459 (270)	1,415 (249)	1,332 (294)	1,334 (285)	1,241 (166)
Total area	52,912 (9,123)	51,813 (9,845)	54,286 (8,702)	55,834 (8,112)	59,091 (6,899)	63,287 (9,153)

*Abdominal wall includes external and internal oblique and transverse abdominis.

**Erector mass includes longissimus thoracis, iliocostalis lumborum, and multifidus.

Muscles	VERTEBRAL LEVEL						
	T11	T10	T9	T8	T7	T6	T5
R rectus abdominis							
L rectus abdominis							
R external oblique							
L external oblique							
R internal oblique							
L internal oblique							
R trans. abdominis							
L trans. abdominis							
R abdominal wall*							
L abdominal wall*							
R longissimus thor.	938 (49)						
L longissimus thor.	938 (21)						
R iliocostalis lumb.	556 (234)						
L iliocostalis lumb.	551 (170)						
R multifidus	331 (89)	351 (90)	312 (97)				
L multifidus	327 (80)	353 (53)	355 (73)				
R latissimus dorsi	1,254 (281)	1,368 (330)	1,458 (269)	1,581 (159)	1,764 (289)	1,876 (432)	2,477 (246)
L latissimus dorsi	1,102 (316)	1,239 (257)	1,417 (293)	1,582 (281)	1,697 (189)	2,013 (422)	2,596 (721)
R erector mass**	1,832 (282)	1,690 (210)	1,413 (304)	1,049 (201)	842 (165)	777 (189)	743 (70)
L erector mass**	2,041 (285)	1,722 (279)	1,471 (351)	1,129 (100)	879 (114)	779 (95)	675 (76)
R psoas							
L psoas							
R quadratus lumb.							
L quadratus lumb.							
Disc area	1,133 (124)	1,015 (125)	933 (112)	798 (91)	797 (104)	741 (80)	671 (82)
Total area	59,249 (7,272)	61,051 (7,570)	61,732 (6,960)	65,794 (5,254)	67,782 (3,982)	66,410 (2,372)	69,337 (2,233)

*Abdominal wall includes external and internal oblique and transverse abdominis.

**Erector mass includes longissimus thoracis, iliocostalis lumborum, and multifidus.

**A.2 Raw Lateral Distances (mm) Between Muscle Centroids and Intervertebral Disc Centroid
(Standard Deviation in Parentheses)**

Muscles	VERTEBRAL LEVEL					
	L5	L4	L3	L2	L1	T12
R rectus abdominis	32 (5)	38 (7)	43 (7)	46 (8)	37 (8)	
L rectus abdominis	-33 (6)	-36 (7)	-38 (8)	-43 (7)	-35 (17)	
R external oblique		125 (13)	130 (10)	140 (5)		
L external oblique		-120 (9)	-125 (9)	-133 (7)		
R internal oblique		109 (11)	116 (8)	123 (9)		
L internal oblique		-103 (9)	-112 (8)	-121 (11)		
R trans. abdominis	99 (1)	108 (11)	122 (9)	117 (9)		
L trans. abdominis	-101 (1)	-101 (9)	-107 (7)	-109 (9)		
R abdominal wall*	102 (8)	113 (12)	119 (8)	123 (9)		
L abdominal wall*	-102 (9)	-115 (14)	-114 (7)	-120 (9)		
R longissimus thor.			22 (4)	32 (2)	32 (6)	30 (2)
L longissimus thor.			-19 (5)	-30 (6)	-37 (12)	-34 (4)
R iliocostalis lumb.			52 (4)	58 (4)	68 (10)	65 (7)
L iliocostalis lumb.			-48 (6)	-60 (10)	-65 (9)	-67 (7)
R multifidus			11 (1)	13 (4)	13 (3)	10 (3)
L multifidus			-14 (7)	-12 (3)	-11 (3)	-11 (2)
R latissimus dorsi			102 (8)	108 (8)	122 (12)	129 (10)
L latissimus dorsi			-104 (15)	-107 (9)	-117 (11)	-128 (7)
R erector mass**	22 (6)	34 (7)	40 (4)	42 (4)	44 (5)	42 (3)
L erector mass**	-21 (5)	-33 (6)	-38 (5)	-41 (6)	-41 (7)	-40 (4)
R psoas	54 (4)	50 (3)	44 (3)	39 (2)	32 (3)	32 (3)
L psoas	-54 (4)	-48 (4)	-42 (3)	-38 (3)	-31 (3)	-32 (2)
R quadratus lumb.		81 (5)	75 (6)	63 (5)	46 (6)	46 (11)
L quadratus lumb.		-78 (12)	-73 (4)	-64 (5)	-50 (6)	-47 (5)
Total area	0 (2)	1 (3)	-2 (4)	-1 (3)	-1 (4)	0 (3)

*Abdominal wall includes external and internal oblique and transverse abdominis.

**Erector mass includes longissimus thoracis, iliocostalis lumborum, and multifidus.

Muscles	VERTEBRAL LEVEL						
	T11	T10	T9	T8	T7	T6	T5
R rectus abdominis							
L rectus abdominis							
R external oblique							
L external oblique							
R internal oblique							
L internal oblique							
R trans. abdominis							
L trans. abdominis							
R abdominal wall*							
L abdominal wall*							
R longissimus thor.							
L longissimus thor.	29 (1)						
R iliocostalis lumb.	-36 (7)						
L iliocostalis lumb.	61 (4)						
R multifidus	-67 (11)						
L multifidus	8 (2)	11 (2)	12 (2)				
R latissimus dorsi	-12 (2)	-12 (2)	-15 (10)				
L latissimus dorsi	129 (9)	140 (9)	141 (8)	145 (7)	146 (7)	153 (7)	153 (4)
R erector mass**	-129 (10)	-137 (9)	-139 (8)	-143 (6)	-147 (10)	-153 (5)	-151 (5)
L erector mass**	34 (4)	34 (4)	32 (4)	31 (7)	30 (4)	25 (5)	27 (2)
R psoas	-40 (3)	-36 (3)	-35 (4)	-33 (6)	-31 (2)	-29 (3)	-27 (6)
L psoas							
R quadratus lumb.							
L quadratus lumb.							
Total area	1 (3)	0 (2)	0 (2)	2 (1)	2 (1)	1 (3)	2 (3)

*Abdominal wall includes external and internal oblique and transverse abdominis.

**Erector mass includes longissimus thoracis, iliocostalis lumborum, and multifidus.

A.3 Raw Anteroposterior Distances (mm) Between Muscle Centroids and Intervertebral Disc Centroid (Standard Deviation in Parentheses)

Muscles	VERTEBRAL LEVEL					
	L5	L4	L3	L2	L1	T12
R rectus abdominis	81 (16)	73 (14)	79 (13)	91 (14)	109 (8)	
L rectus abdominis	80 (15)	73 (14)	81 (14)	92 (14)	112 (6)	
R external oblique		35 (10)	20 (14)	28 (12)		
L external oblique		32 (18)	19 (11)	28 (11)		
R internal oblique		41 (12)	25 (9)	36 (17)		
L internal oblique		41 (17)	26 (12)	40 (16)		
R trans. abdominis	55 (0)	28 (11)	22 (11)	36 (6)		
L trans. abdominis	50 (5)	30 (14)	23 (10)	44 (5)		
R abdominal wall*	58 (16)	31 (12)	17 (12)	30 (15)		
L abdominal wall*	59 (17)	32 (13)	20 (12)	31 (11)		
R longissimus thor.			-61 (6)	-62 (7)	-60 (7)	-60 (6)
L longissimus thor.			-61 (5)	-63 (6)	-60 (7)	-59 (8)
R iliocostalis lumb.			-57 (7)	-61 (7)	-62 (5)	-59 (7)
L iliocostalis lumb.			-57 (7)	-61 (6)	-61 (4)	-58 (6)
R multifidus			-55 (7)	-55 (6)	-52 (6)	-51 (3)
L multifidus			-53 (7)	-56 (5)	-51 (5)	-50 (3)
R latissimus dorsi			-45 (16)	-47 (12)	-47 (10)	-39 (8)
L latissimus dorsi			-43 (17)	-46 (10)	-46 (7)	-37 (8)
R erector mass**	-64 (6)	-61 (5)	-61 (5)	-61 (5)	-59 (5)	-56 (5)
L erector mass**	-63 (5)	-61 (5)	-61 (5)	-62 (5)	-60 (4)	-57 (5)
R psoas	18 (9)	1 (5)	-7 (5)	-9 (5)	-11 (6)	-14 (2)
L psoas	19 (8)	2 (4)	-6 (4)	-8 (2)	-11 (4)	-11 (1)
R quadratus lumb.		-36 (9)	-37 (6)	-37 (6)	-35 (4)	-31 (6)
L quadratus lumb.		-31 (5)	-34 (6)	-36 (5)	-34 (4)	-32 (8)
Total area	1 (10)	-2 (9)	1 (8)	9 (8)	18 (5)	24 (7)

*Abdominal wall includes external and internal oblique and transverse abdominis.

**Erector mass includes longissimus thoracis, iliocostalis lumborum, and multifidus.

	VERTEBRAL LEVEL						
Muscles	T11	T10	T9	T8	T7	T6	T5
R rectus abdominis							
L rectus abdominis							
R external oblique							
L external oblique							
R internal oblique							
L internal oblique							
R trans. abdominis							
L trans. abdominis							
R abdominal wall*							
L abdominal wall*							
R longissimus thor.	-56 (4)						
L longissimus thor.	-52 (4)						
R iliocostalis lumb.	-57 (1)						
L iliocostalis lumb.	-56 (2)						
R multifidus	-47 (5)	-49 (4)	-48 (2)				
L multifidus	-47 (5)	-47 (3)	-47 (2)				
R latissimus dorsi	-32 (7)	-24 (7)	-22 (7)	-18 (9)	-17 (6)	-12 (3)	-17 (5)
L latissimus dorsi	-28 (9)	-23 (7)	-19 (7)	-17 (7)	-15 (8)	-11 (7)	-19 (3)
R erector mass**	-54 (4)	-54 (4)	-52 (4)	-52 (3)	-52 (4)	-47 (4)	-50 (3)
L erector mass**	-52 (4)	-52 (4)	-51 (4)	-51 (3)	-51 (4)	-46 (5)	-50 (3)
R psoas							
L psoas							
R quadratus lumb.							
L quadratus lumb.							
Total area	29 (8)	30 (9)	32 (7)	36 (7)	37 (6)	32 (10)	34 (5)

Glossary

- anisotropic**—A description of a material whose behavior depends on the direction of the applied load. For example, adult bone is stronger in compression than in tension.
- antagonist**—A muscle that opposes the function of another in terms of torque generation but that may be necessary to ensure a stable joint.
- breaking strain**—The strain at which tissue fails.
- buckle**—An abnormally large rotation at a single joint, resulting in very large passive tissue forces. The spine may experience several modes of unstable behavior. The column may buckle under compressive loads if sufficient stiffness is not present in the rotational degrees of freedom.
- dermatome**—An area of skin supplied with afferent nerve fibers by a single spinal nerve root.
- discogenic** —Relating to or originating from the disc (e.g., discogenic pain or discogenic disorders).
- eigenvalue**—The element of the Hessian matrix that, if less than zero, indicates the potential for unstable behavior.
- eigenvector**—The mode and site of unstable behavior, and the deficit that allowed the instability to develop.
- electromyography**—The measurement of the electromyogram, electrical activity associated with muscle contraction.
- engram**—The encoded representation of a movement, which resides in the motor cortex or the spinal cord. Most repeated movements arise from "running the tape," or engram. The movement may then be modified by reflexes if the normal pattern is disturbed.
- etiology**—The analysis and description of the cause of injury or disease.
- failure tolerance**—The load at which a tissue fails.
- lordosis**—The natural curvature of the neutral lumbar spine; hypolordosis occurs with a flexed lumbar spine; hyperlordosis occurs with an extended lumbar spine.
- moment**—Moment of force; occurs when a force is applied through a distance about an axis to create rotational effort.
- moment arm**—The perpendicular distance between the axis of rotation and a force, which creates a moment.

myoelectric—Of or relating to an electrical signal obtained from contracting muscle.

myotome—An electrical signal obtained from contracting muscle.

neural arch—The structure that encloses the spinal cord at each vertebra; includes the pedicles and laminae.

neutral lordosis—When the lumbar spine is neither flexed nor extended.

NIOSH—National Institute for Occupational Safety and Health (United States).

nociception—The sensation of pain with a nociceptor (pain-sensing organ).

odds ratio—The increase in the odds of an event occurring.

osteophytes—Bony spurs that grow on the stress points of a vertebra; usually indicative of an unstable joint.

palpation—Using the hands to feel for spinal motion or pathology.

pars interarticularis —The part of the vertebra that connects the facet articulating surface to the neural arch and often is the site of fracture.

pathomechanics —Mechanics associated with pathology.

Poisson's effect—A material property of an elastic sheet in which strain longitudinally produces a proportional lateral strain.

provocative , or functional, diagnosis—A diagnosis based on tasks that exacerbate symptoms.

provocative testing—Provoking tissues with movement or direct load to elicit pain.

psychophysical—A description of variables that are associated with psychological or social aspects of a person.

psychosocial—A description of variables that influence behavior that may include psychological traits such as the perception of control over a task and social traits such as how people are viewed by others.

spondylolisthesis—A fracture of the pars interarticularis that causes an anterior shear displacement of the superior vertebra (usually L4 or L5).

strain—A change in length normalized to rest length (dimensionless ratio) and expressed as a percentage.

subfailure —An applied load that is less than the load required to cause damage if applied once.

synergist—A muscle that assists another in the same functional objective.

tolerance—A load value above which injury occurs. It is modulated by repetition, duration, rest, and so on.

transmissible vector—The transmissible property of any force vector is due

to its having an effect along the line on which it is directed. A perpendicular line can be drawn from the force vector to any body joint to determine the moment arm, or the potential of that force to cause a moment.

viscoelastic—The property of being both elastic and viscous to absorb the shock of loads.

viscosity—Friction as a function of velocity; it always moves in the direction of motion or impending motion.

yield point—The load at which a tissue begins to experience damage.